

DoS

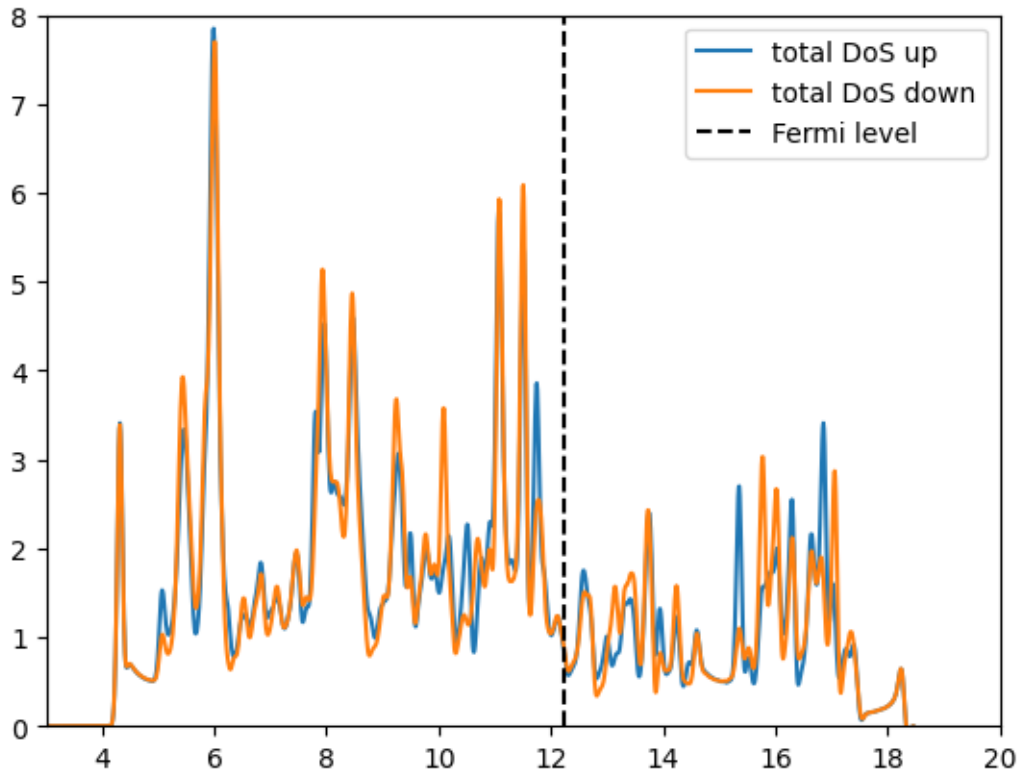
July 4, 2025

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[2]: import numpy as np
import matplotlib.pyplot as plt

[3]: DoS = np.array(np.loadtxt('Ru_pdos.dat.pdos_tot', skiprows=1, usecols=(0, 1, 2, 3, 4), unpack=True, dtype=float))
DoSRu1 = np.array(np.loadtxt('Ru_pdos.dat.pdos_atm#1(Ru1)_wfc#4(d)', skiprows=1, usecols=(0, 1, 2, 3, 4), unpack=True, dtype=float))
DoSRu2 = np.array(np.loadtxt('Ru_pdos.dat.pdos_atm#2(Ru2)_wfc#4(d)', skiprows=1, usecols=(0, 1, 2, 3, 4), unpack=True, dtype=float))
DoS01 = np.array(np.loadtxt('Ru_pdos.dat.pdos_atm#3(0)_wfc#2(p)', skiprows=1, usecols=(0, 1, 2, 3, 4), unpack=True, dtype=float))
DoS02 = np.array(np.loadtxt('Ru_pdos.dat.pdos_atm#4(0)_wfc#2(p)', skiprows=1, usecols=(0, 1, 2, 3, 4), unpack=True, dtype=float))
DoS03 = np.array(np.loadtxt('Ru_pdos.dat.pdos_atm#5(0)_wfc#2(p)', skiprows=1, usecols=(0, 1, 2, 3, 4), unpack=True, dtype=float))
DoS04 = np.array(np.loadtxt('Ru_pdos.dat.pdos_atm#6(0)_wfc#2(p)', skiprows=1, usecols=(0, 1, 2, 3, 4), unpack=True, dtype=float))
DoS = DoS.T
DoSRu1 = DoSRu1.T
DoSRu2 = DoSRu2.T
DoS01 = DoS01.T
DoS02 = DoS02.T
DoS03 = DoS03.T
DoS04 = DoS04.T

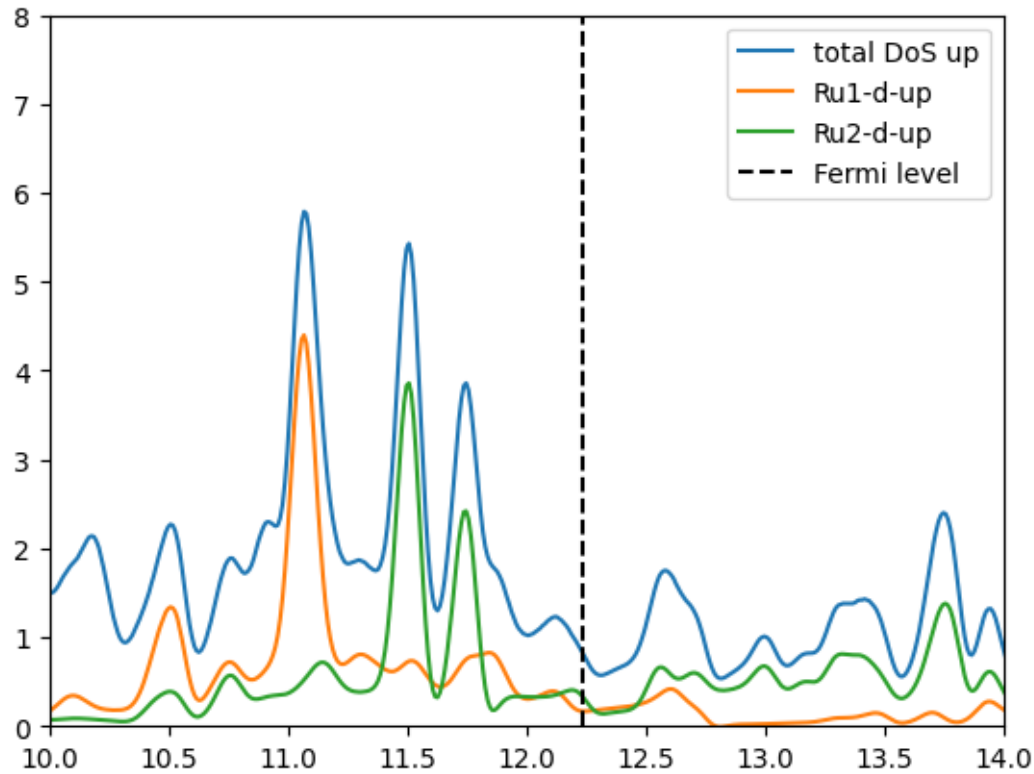
[4]: plt.plot(DoS[:, 0], (DoS[:, 1]), label='total DoS up')
plt.plot(DoS[:, 0], (DoS[:, 2]), label='total DoS down')
plt.axvline(x = 12.2400, label='Fermi level', color='black', linestyle='--')
plt.xlim(3,20)
plt.ylim(0,8)
plt.legend()

[4]: <matplotlib.legend.Legend at 0x22c52889fd0>
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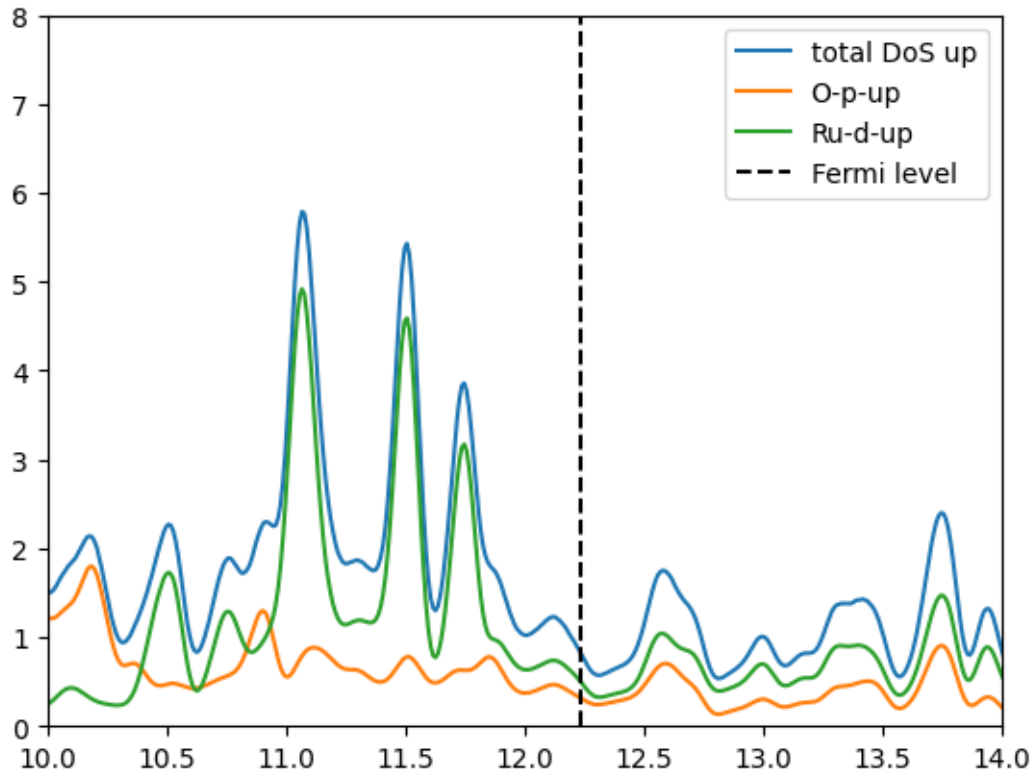
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[5]: plt.plot(DoS[:, 0], (DoS[:, 1]), label='total DoS up')
plt.plot(DoSRu1[:, 0], (DoSRu1[:, 1]), label='Ru1-d-up')
plt.plot(DoSRu2[:, 0], (DoSRu2[:, 1]), label='Ru2-d-up')
plt.axvline(x = 12.2400, label='Fermi level', color='black', linestyle='--')
plt.xlim(10,14)
plt.ylim(0,8)
plt.legend()
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[5]: <matplotlib.legend.Legend at 0x22c54ad8f50>



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[6]: plt.plot(DoS[:, 0], (DoS[:, 1]), label='total DoS up')
DoS0 = DoS01 + DoS02 + DoS03 + DoS04
DoSRu = DoSRu1 + DoSRu2
plt.plot(DoS[:, 0], DoS0[:, 1], label='O-p-up')
plt.plot(DoS[:, 0], DoSRu[:, 1], label='Ru-d-up')
plt.axvline(x = 12.2400, label='Fermi level', color='black', linestyle='--')
plt.xlim(10,14)
plt.ylim(0,8)
plt.legend()
```

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[6]: <matplotlib.legend.Legend at 0x22c54942d50>
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[9]: plt.plot(DoS[:, 0], (DoS[:, 1] + DoS[:, 2]), label='total DoS')
DoS0 = DoS01 + DoS02 + DoS03 + DoS04
DoSRu = DoSRu1 + DoSRu2
plt.plot(DoS[:, 0], DoS0[:, 1] + DoS0[:, 2], label='O-p')
plt.plot(DoS[:, 0], DoSRu[:, 1] + DoSRu[:, 2], label='Ru-d')
plt.axvline(x = 12.2400, label='Fermi level', color='black', linestyle='--')
plt.xlim(10,14)
plt.ylim(0,15)
plt.legend()
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[9]: <matplotlib.legend.Legend at 0x22c54c59950>
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